Final Exam Prepare - 25/9 exam

Chap 1

1. Suggest an appropriate software stack (operating system, web server, database server and programming runtime) for hosting ASP.NET web applications.

1. Software stack (???) for hosting ASP.NET web applications. // Page 3 - 4

- Operating System: Windows server
- Web Server: Internet Information Services (IIS)
- Database server: sql server
- Programming Runtime: .NET 10 (cross-platform)

3. Identify and discuss the role of the 3 core components of MVC web architecture. Provide 2 examples for each.

2. The role of the 3 core components of MVC web architecture. // Page 27

	Model	View	Controller
	Data layer	Presentation layer	Application logic layer
	Represent data of the application	Template files the application users	Handle incoming requests
E.g.	Database, entity classes	HTML, CSS	C# controller classes, action method

3. Advantages of MVC web architecture // Page 28 - 31

Separation of concerns
- Each MVC components focuses on specify tasks (modular)

Task distribution
- Making it easier to distribute tasks among developers, database administrators and designers
  - Developers: focus on controllers
  - Database administrators: focus on models
  - Designers: focus on views

Reusable
- MVC application is modular

4. Advantages of using the ASP.NET platform for developing web applications // Page 8-9

Existing software environment
- Many companies may already use Windows-based software

Customer support
- Windows-based software are usually paid

Expandability and scalability
- Companies and businesses will grow and expand

5. Why URL routing is necessary for ASP.NET MVC web applications. // Page 50-51

- URL routing engine maps incoming requests (URLs) against the routing rules

6. URL Mapping Analysis // Page 52-54 - read slides

Following the default routing pattern {controller}/{action}/{id}

Example Question 1- Discuss in words how the following URL is mapped:
https://www.abc.com/Product/Update/P001?quantity=123
Ans:
Controller: Product.
Action Method: Update.
ID Parameter: P001.
Query String Parameter: quantity (value is 123)

Example Question 2 - Write the URL to access the StudentController and Detail action method. Pass S001 for id and 2025 for year.
Ans:
The URL would be: https://www.abc.com/Student/Detail/S001?year=2025

Code Format

```
public class _____ : Controller
{
  public IActionResult _____ ( _____ , _____ )
  {
    ...
  }
}
```

Example Question 3 - Rewrite the following code filling in the missing parts to map to:
https://www.abc.com/Product/Update/P001?quantity=123
Ans:

```
public class ProductController : Controller
{
        public IActionResult Update (string id, int quantity)
        {
        }
```

}

Chap 2

1. Advantages of using reusable layout in ASP.NET MVC web applications? // Page 3-4

- Provide users with consistent experience as they navigate from page to page.
- Layout reduces duplicate code in views.

3. Differentiate between tag helpers and HTML helpers. (Page 39)

Tag helpers	HTML helpers
Written as HTML elements with custom non-standard attributes	Written as C# methods. Work like function calls with parameters
Codes are clean, concise and easier to understand	Codes are usually more verbose and slightly complicated
Codes are more readable as they resemble typical HTML syntax	Codes are less readable as HTML and C# codes are mixed together

Chap 3

2. Define unobtrusive JavaScript. (Page 5)
- Separation of JavaScript (behavior) from HTML (markup)
- Avoid the use of inline JavaScript for event handling

4. Discuss the advantages of writing unobtrusive JavaScript.
(Page 8-10)
Modularity
- Making JS modules plug-able and reusable
Task distribution
- Web designers can work on HTML and CSS without worrying how to program JS code
Maintainability
- making the code easier to read (less messy)

6. Discuss AJAX programming technique and its advantages.
(Page 20)
AJAX = Asynchronous JavaScript and XML
- Handle HTTP request and response using JavaScript running in the background
- Allow form submission and content update without the need of reloading whole page

Advantages: Improve responsiveness and user experience

Chap 4

1. What is object-relational mapping (ORM)? Identify how relational model (table, column, record and relationship) is mapped to object model. (Page 3)

ORM is a process of converting data in a relational database into object-oriented representations.

Mapping from relational model to object model:

Relational Model		Object Model
Table	map to	Class
Column		Property
Relationship		Association
Record (Row)		Object

4. Differentiate between database-first and code-first database development. (Page 11)

Database-First	Code-First
Develop database first. Then generate entity classes automatically	Develop entity classes first. Then generate database automatically
Popular for database administrators who focus on database development	Popular for programmers who focus on application development
Suitable when working with existing database	Suitable when the database has not been created

Chap 5

4. Differentiate between GET and POST request methods. Classify the SELECT, INSERT, UPDATE and DELETE operations to either GET or POST request methods respectively. (Page 44)

GET	POST
To retrieve resource from web server	To create, modify or remove resource from web server.
For action that does not alter server state	For action that alters server state
No side effect	Has side effect
Example: SELECT operation	Example: INSERT, UPDATE and DELETE operations. (Choose any 1)

5. Identify if the GET or POST request method is suitable for each of the following operations. State the reason clearly: (Page 46) #TODO

GET是get data/找出来的意思
Ans Format: GET. Because it does not alter server state or data.
POST是类似upload data的意思
Ans Format: POST. Because it alter server state or data.

Select a student record
- GET. Because it does not alter server state or data.

Insert a student record
- POST. Because it alter server state or data.

Update a student record
- GET. Because it does not alter server state or data.

Delete a student record
- POST. Because it alter server state or data.
Search a list of student records by name
- GET. Because it does not alter server state or data.
Sort a list of student records by gender
- GET. Because it does not alter server state or data.
Download a student profile photo
- GET. Because it does not alter server state or data.
Upload a student profile photo
- POST. Because it alter server state or data.

6&7. Examine the advantages & disadvantages of using the ExecuteUpdate() and ExecuteDelete() methods. (Page 71 & 72)

Advantages	Disadvantages
Allow update or delete one or multiple records without pre-select the targeted records.	The syntax is more SQL-oriented rather than object-oriented.
The underlying SQL query is generated and sent directly to the database for execution.	The syntax may be less readable for UPDATE operation.
Less memory usage	-

Chap 6

3. Differentiate between client-side and server-side input validations. Then recommend which one should be implemented into modern web systems. (Page 21-23)

Recommend client-side input validations

client-side	server-side
Program using JS. Run at browser (locally).	Program using C#. Run at the server (remotely).
However, incapable to perform complex validation logics.	However, involve roundtrip, thus is less responsive.
Insecure, as JS can be turned off.	Secure, server-side input validations always run.

4. Differentiate between strongly-typed models and ViewBag properties. (Page 58)

strongly-typed models	ViewBag properties
A view can only accept one model.	Capable to pass multiple data to view.
Support auto code completion and design-time debugging.	Does not support auto code completion and design-time debugging.
It is usually used to hold main (primary) data of the page.	It is usually used to hold sub (secondary) data of the page.

Chap 7

3. Identify and discuss the THREE (3) image processing techniques. Justify why each technique is important for web systems. (Page27-29)
Image Cropping
- Cut or drop the excessive portions of the image.
Image Resizing
- Reduce the dimension of the image.
(example: width and height)
Image Type Conversion
- Convert images of different types to a single consistent type
(example: BMP,PNG□JPG).

Chap 8

1. Differentiate between authentication and authorization. For each of them, provide a use case example of how it can be implemented in a web system. (Page 3)

	authentication (身份验证)	authorization（授权 ）
	The process of ascertaining the identity of a user. Ensure if the user is really the one who he claims to be.	The process of allowing or denying a user from accessing protected pages, functions or resources.
Example	Logging in with a password or OTP.	An admin editing a post while a guest can only view it.

3. Refer to the codes given below. Analyse and discuss the effect of Line 01, 04 and 07: (Page 7)

public IActionResult Index() { ... }
- Everyone can access to this action method, even without login

```
01 [Authorize]
02 public IActionResult Page1() {...}
03
04 [Authorize(Roles = "Admin")]
05 public IActionResult Page2() {...}
06
07 [Authorize(Roles = "Admin, Member")]
08 public IActionResult Page3() {...}
```

01 line
- Requires the user to be authenticated, any logged-in user can access

04 line
- Only users with the Admin role can access. Others will be denied.

07 line
- Only users with Admin or Member roles can access. Users with neither role are denied.

6. Investigate how the following web optimization techniques work: (Page 28-30)

Bundling
- Combine multiple smaller JS/CSS files together as a single larger file
- Making multiple HTTP requests can slow down web app loading time

Minification
- Reduce JS/CSS file size by removing unnecessary characters
- Also rename identifiers to shorter names